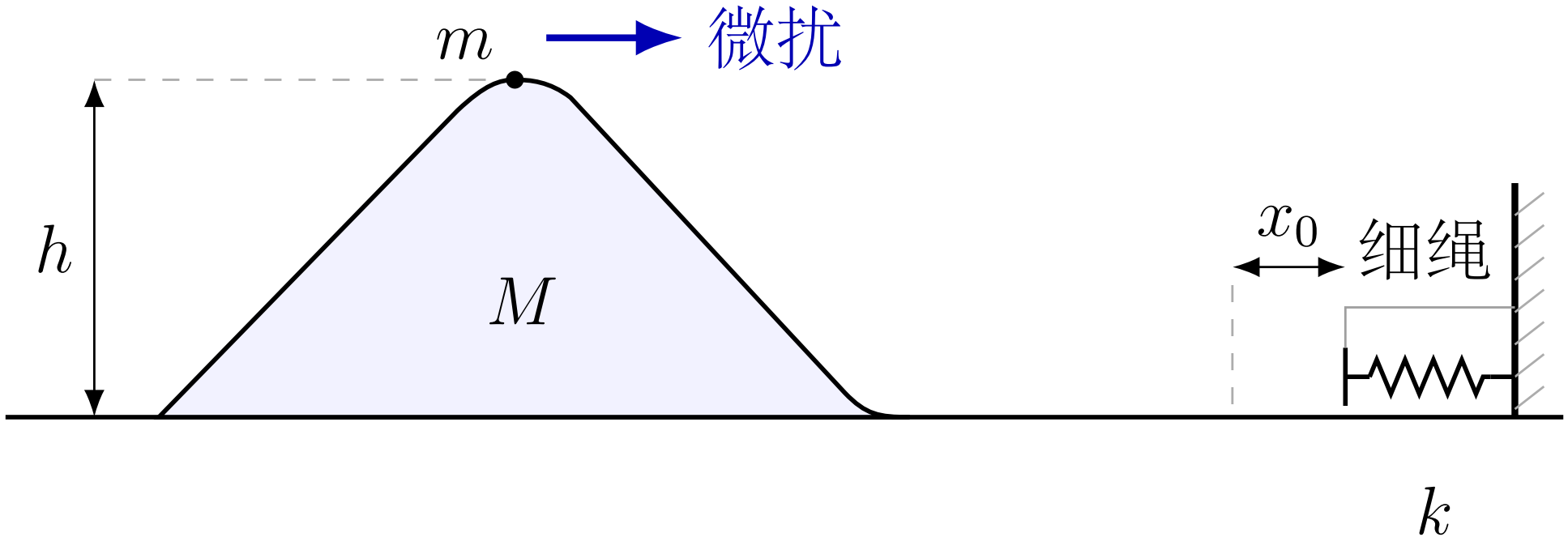
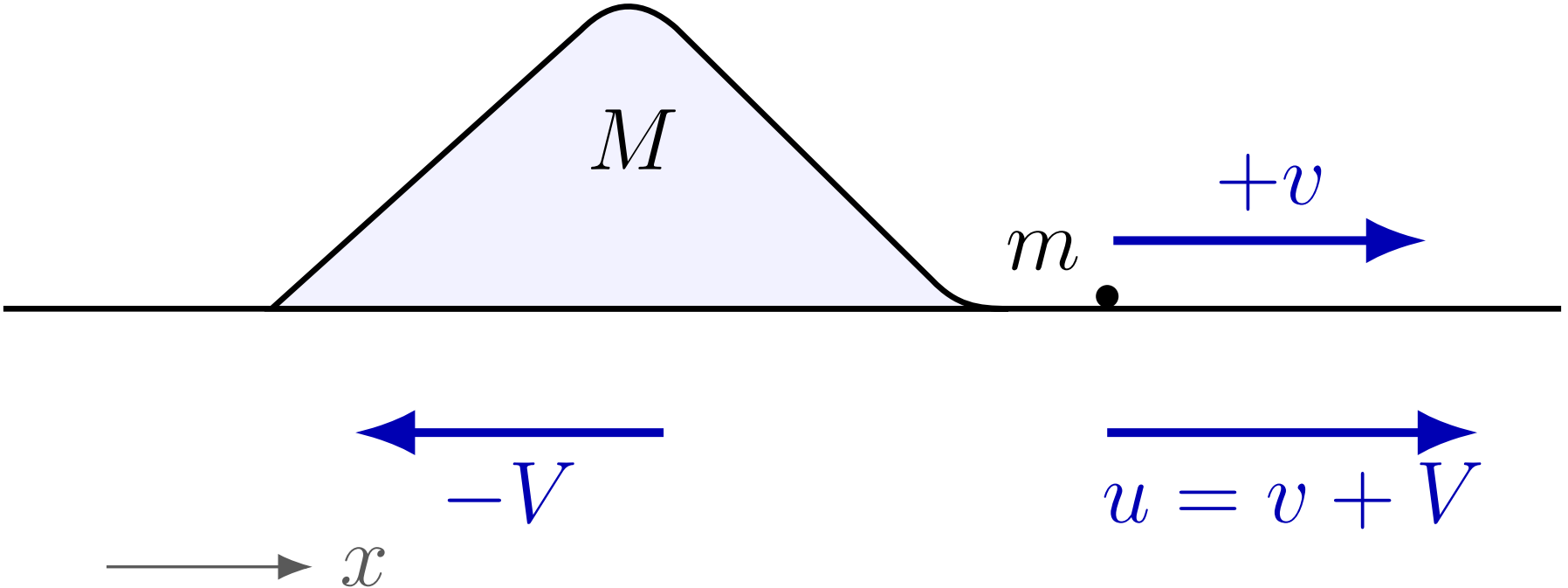
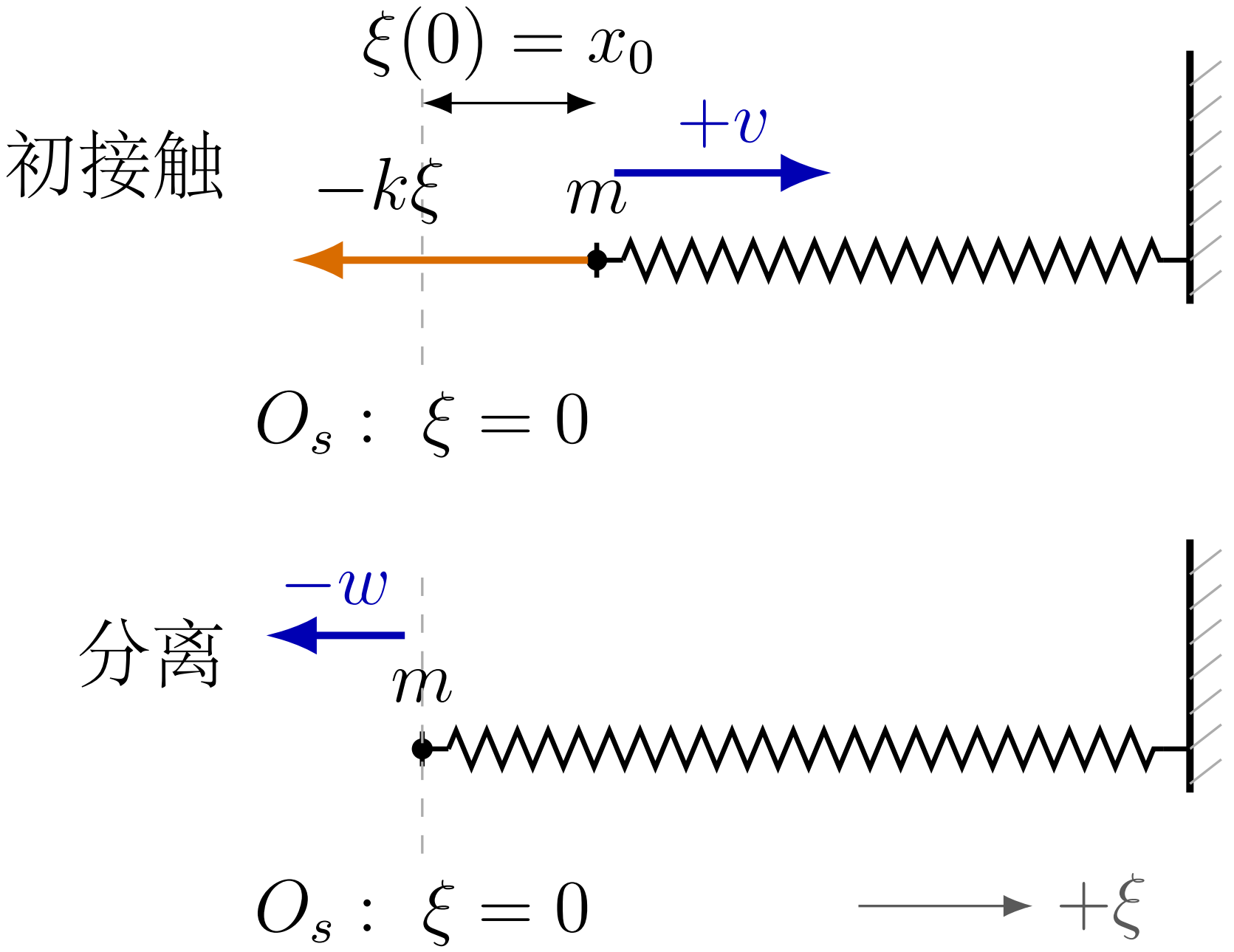
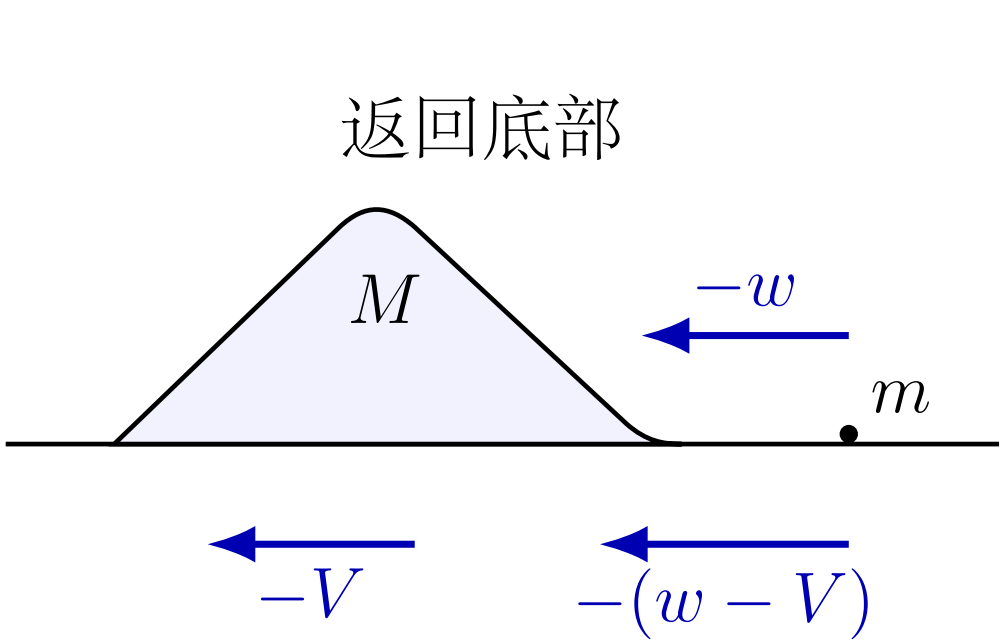
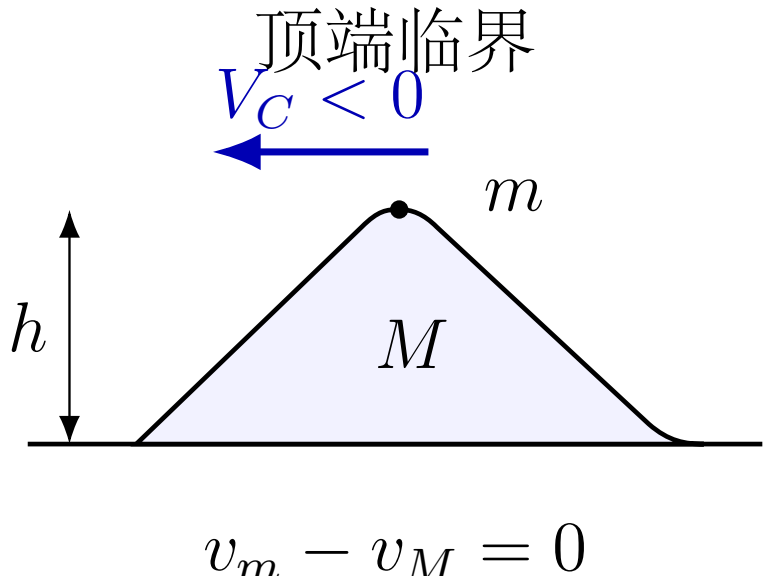
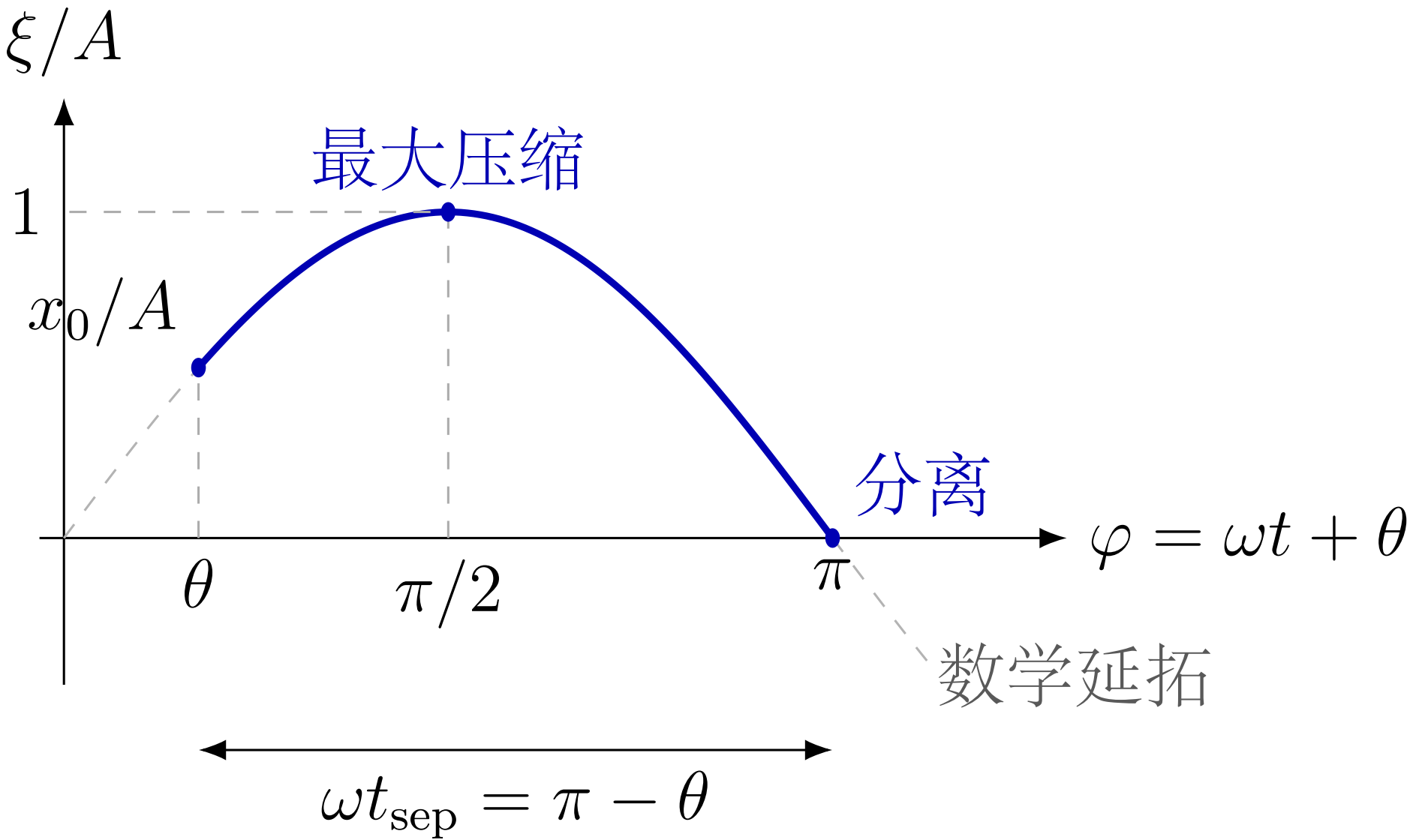

m
微扰
h
M
x_0
细绳
k


M
$+v$
m
$-V$
$u = v + V$
x


$\xi(0) = x_0$
初接触
$+v$
$-k\xi$
m
$O_s:\ \xi = 0$
$-w$
分离
m
$O_s:\ \xi = 0$
$+\xi$


返回底部
M
−w
m
−V
−(w − V)


顶端临界
$V_C < 0$
m
h
M
$v_m - v_M = 0$


ξ/A
最大压缩
1
x_0/A
分离
$\varphi = \omega t + \theta$
θ
$\pi/2$
π
数学延拓
$\omega t_{\mathrm{sep}} = \pi - \theta$